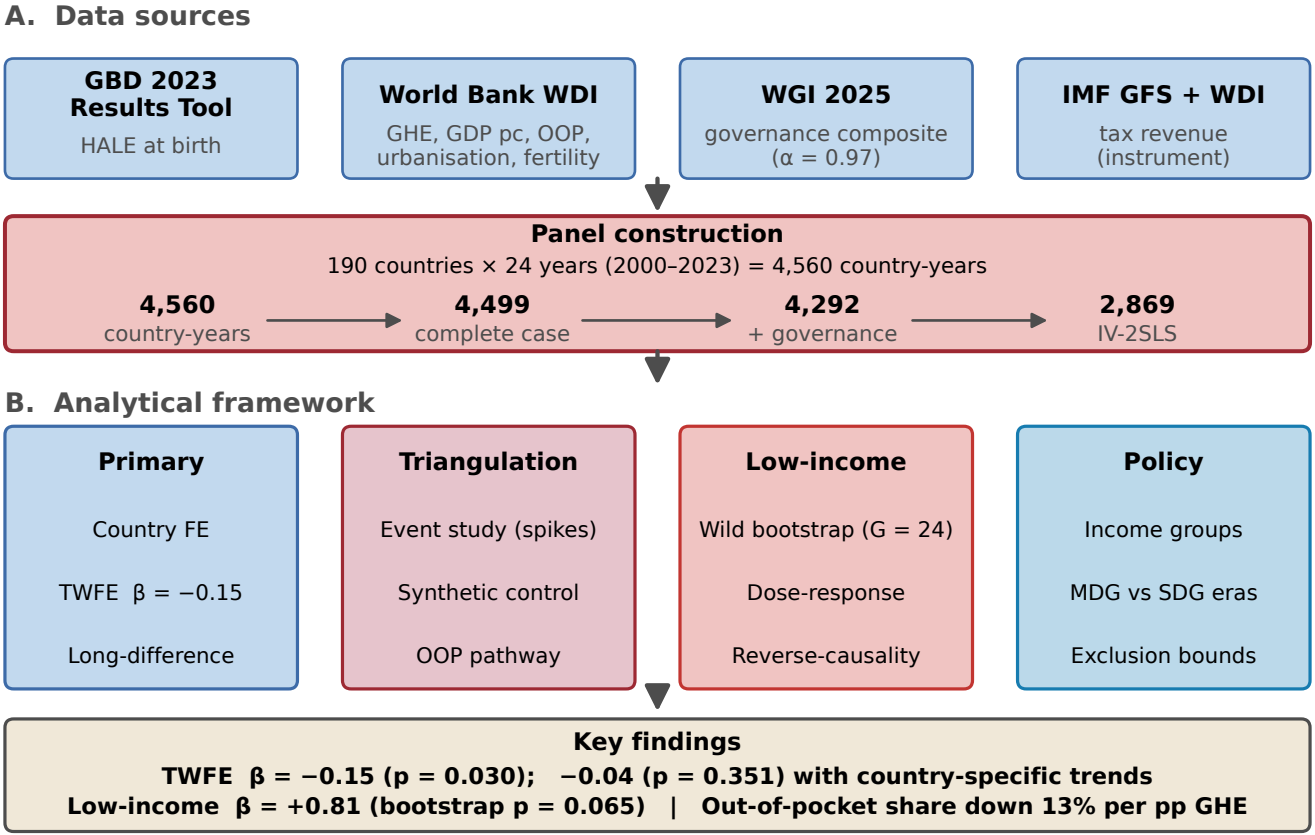
A. Data sources
GBD 2023 Results Tool
HALE at birth
World Bank WDI
GHE, GDP pc, OOP, urbanisation, fertility
WGI 2025
governance composite (α = 0.97)
IMF GFS + WDI
tax revenue (instrument)
Panel construction
190 countries × 24 years (2000–2023) = 4,560 country-years
4,560
country-years
4,499
complete case
4,292
+ governance
2,869
IV-2SLS
B. Analytical framework
Primary
Country FE
TWFE β = −0.15
Long-difference
Triangulation
Event study (spikes)
Synthetic control
OOP pathway
Low-income
Wild bootstrap (G = 24)
Dose-response
Reverse-causality
Policy
Income groups
MDG vs SDG eras
Exclusion bounds
Key findings
TWFE β = −0.15 (p = 0.030); −0.04 (p = 0.351) with country-specific trends
Low-income β = +0.81 (bootstrap p = 0.065) | Out-of-pocket share down 13% per pp GHE